

IPW deep-dive (Hájek and Horvitz–Thompson)

Overview

Inverse-probability weighting (IPW) is the simplest of the single-robust causal estimators. This vignette shows the building blocks that MORIE exposes: the Horvitz–Thompson and the Hájek-stabilised IPW estimators, propensity-score modelling, and weight-trimming diagnostics.

Setting up

```
library(morie)
set.seed(2026)
n <- 500
X1 <- rnorm(n)
X2 <- rnorm(n)
ps_true <- plogis(0.4 * X1 - 0.3 * X2)
treat <- as.integer(ps_true > runif(n))
y <- 1.0 * treat + 0.6 * X1 - 0.2 * X2 + rnorm(n, sd = 0.5)
df <- data.frame(y = y, treat = treat, X1 = X1, X2 = X2)
```

Estimating propensities

The `estimate_ate()` machinery fits a logistic propensity model internally and returns the IPW estimate by default. To inspect the propensities, set `propensity_col` after fitting them:

```
ps_fit <- glm(treat ~ X1 + X2, family = binomial(), data = df)
df$ps <- predict(ps_fit, type = "response")
summary(df$ps)
#>      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
#>  0.1604  0.3727  0.4549  0.4580  0.5374  0.7422
```

Hájek-stabilised IPW

`estimate_ate()` defaults to the Hájek estimator, which divides each weighted sum by the corresponding sum of weights. This stabilises the estimator under finite samples even when the propensity tails are heavy:

```
ate_hajek <- estimate_ate(df, treatment = "treat", outcome = "y", covariates = c("X1", "X2"),
                          propensity_col = "ps")
ate_hajek$estimate
#> NULL
ate_hajek$se
#> [1] 0.04755316
```

Weight diagnostics

In practice, IPW is sensitive to extreme propensities. Two common diagnostics:

```
# Effective sample size after weighting
ess <- effective_sample_size(1 / df$ps)
ess
#> [1] 458.4318

# Range of weights (extreme means trimming)
range(1 / df$ps)
#> [1] 1.347363 6.234294
```

If the effective sample size collapses dramatically, the analysis should consider:

- Trimming propensities to a sensible interval (e.g. [0.05, 0.95])
- Switching to a doubly-robust estimator (`estimate_aipw()`)
- Adding more covariates to better separate the treatment groups

AIPW for protection against IPW failure

```
aipw <- estimate_aipw(df, treatment = "treat", outcome = "y", covariates = c("X1", "X2"))
aipw$estimate
#> NULL
```

When propensities are well-behaved, IPW and AIPW should agree to within Monte Carlo noise. Disagreement is informative: it suggests either model misspecification or a fragile propensity model.

Where to go next

- The `causal-inference` vignette covers ATT / ATC / CATE / GATE.
- The `survey-weighted` vignette covers IPW under complex-sample designs (when survey weights and propensities both apply).